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USTC sets new world record in perovskite solar cell efficiency

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UNIVERSITY OF SCIENCE AND TECHNOLOGY OF CHINA

A research team led by Prof. XU Jixian from the University of Science and Technology of China (USTC) has once again pushed the boundaries of solar cell technology. On July 3rd, the prestigious *Solar Cell Efficiency Tables* published Version 64, in which they announce a new world record for perovskite solar cell performance set by Professor Xu's team, with a certified stable efficiency of 26.7%.

The *Solar Cell Efficiency Tables*, published jointly by the Australian Centre for Advanced Photovoltaics alongside scientists from the United States, Japan, Italy, and other countries, has a nearly 30-year history. Its objective assessment of solar cell efficiencies is widely recognized in the international photovoltaic academic and industrial communities.

This marks the third consecutive year that Prof. XU's team has updated the world record for perovskite solar cell efficiency, following their achievements in 2022 (Version 60) and 2023 (Version 63). In 2023, the team set a groundbreaking certified efficiency of 26.1% for their inverted perovskite solar cell, surpassing the 26% efficiency milestone and breaking the dominance of conventional solar cells in world records. The consistent breakthroughs by Prof. Xu's team highlight their relentless pursuit of innovation in photovoltaics. Their latest achievement, the result of continuous and rigorous efforts, represents another leading breakthrough and significantly advances the development of tandem solar cells.

The inclusion of Prof. XU's achievement in these tables underscores USTC's leading capabilities in advanced

Classification	Efficiency (%)	Area (cm ²)	J _{sc} (mA/cm ²)	V _{oc} (V)	Fill factor (%)	Test centre	Description
Perovskite (thin-film)	26.7 ± 0.4 ^{***}	0.0119 (64)	1.73	26.49 ^{***}	84.5	NREL (J-28)	USTC ¹ 中国科大 徐集贤团队
Organic (thin-film)	19.2 ± 0.3 ^{***}	0.0326 (64)	0.9135	26.61 ^{***}	79.0	NREL (J-28)	SJTU ¹²
Dye sensitized	13.0 ± 0.4 ^{***}	0.1155 (64)	1.0396	15.50 ^{***}	80.4	FNG (SEI (50/20))	EPFL ¹³

IMAGE:

USTC ACHIEVED 26.7% EFFICIENCY FOR PEROVSKITE SOLAR CELLS.

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photovoltaic research. This breakthrough is expected to drive further innovation and application in the renewable energy industry, contributing to the transition to more sustainable energy sources.

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